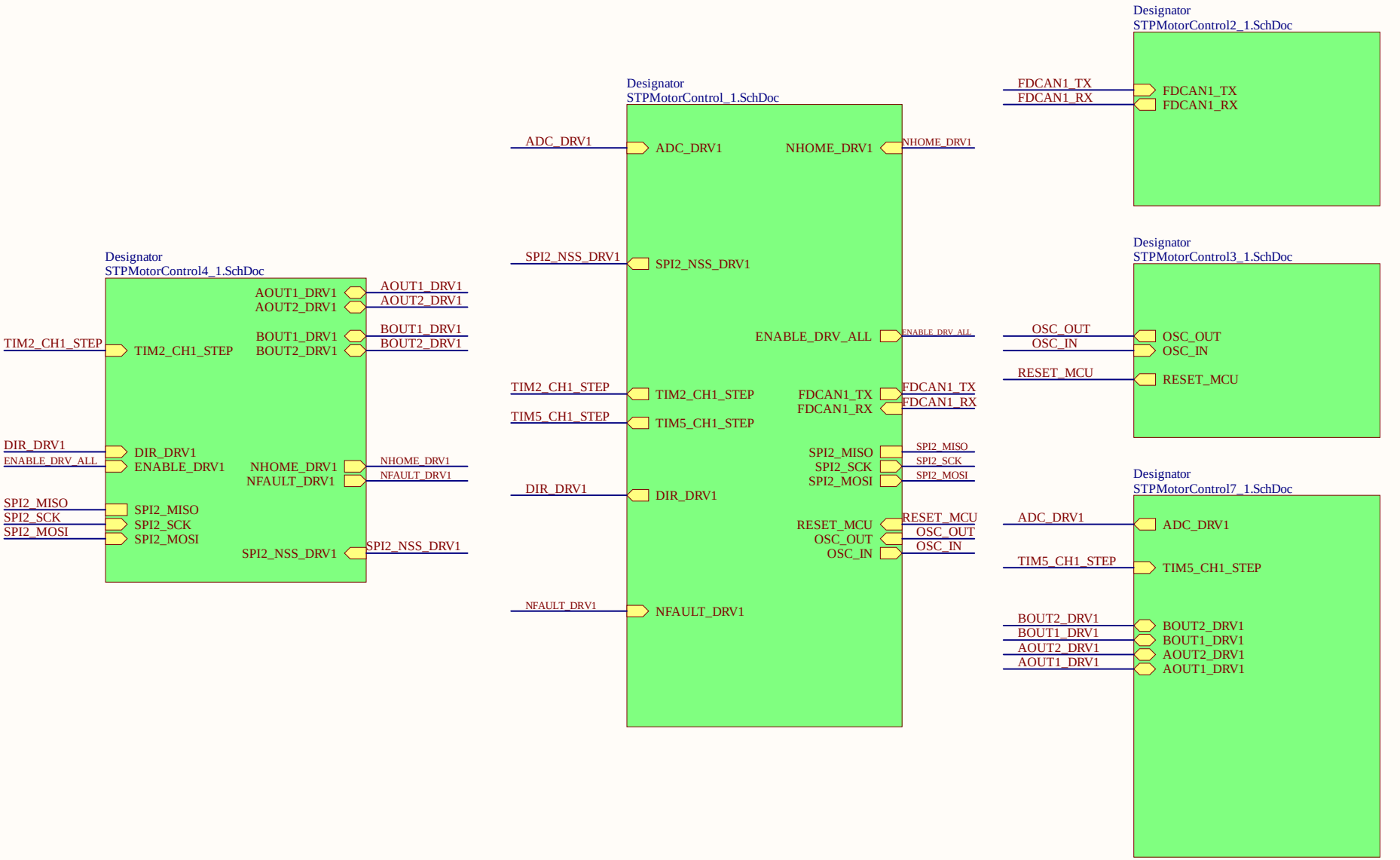
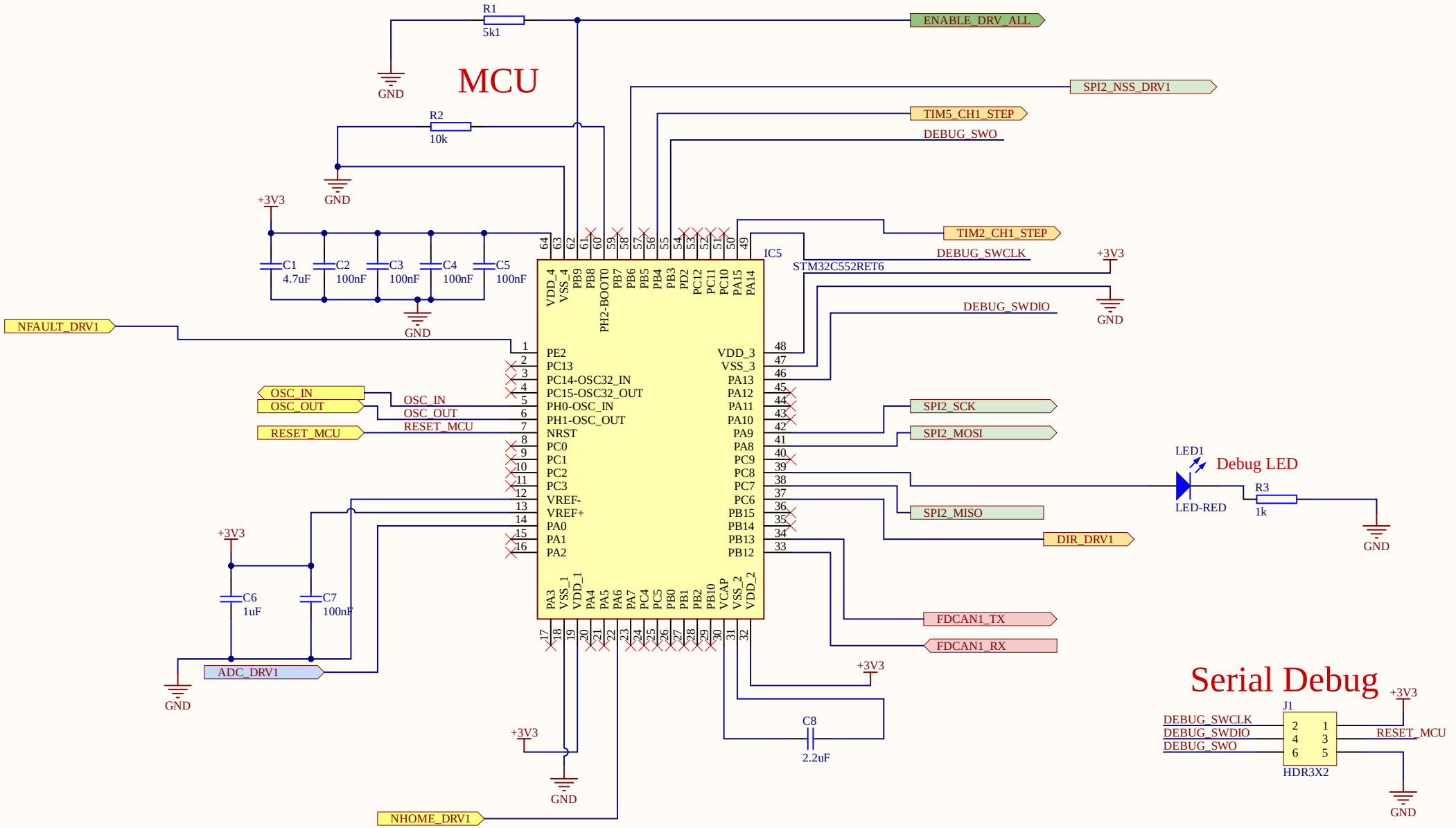
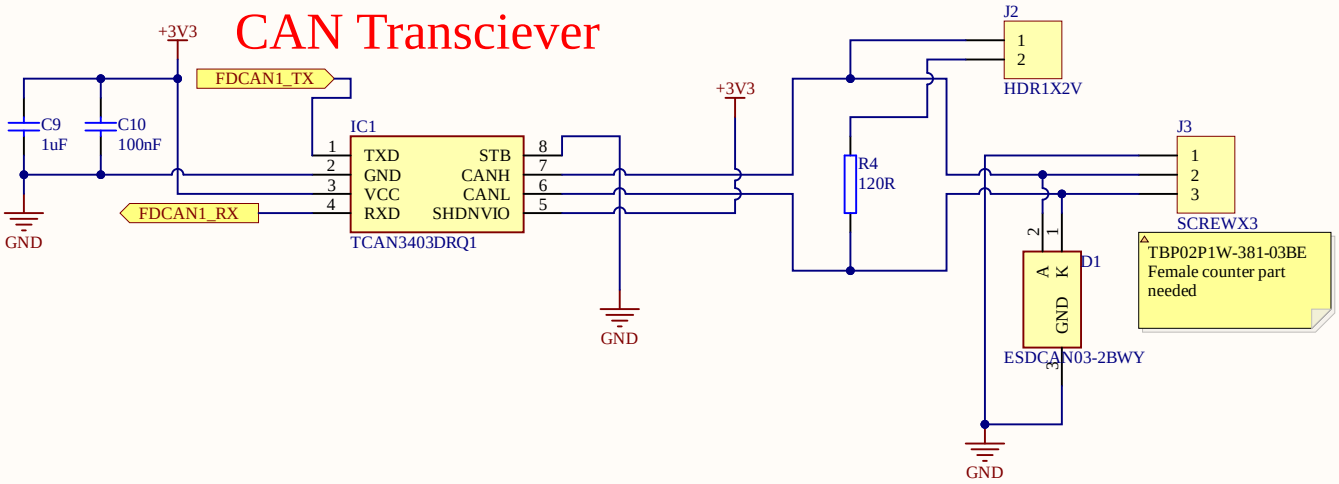


Title	Gimbal Warning Projector Controller	
Size: A4	Number: 1	Revision:
Date: 10/20/2026		Sheet of 1/8
File: C:\Users\..\STPMotorConnections.SchDoc		Drawn By: Mohamed Hanan

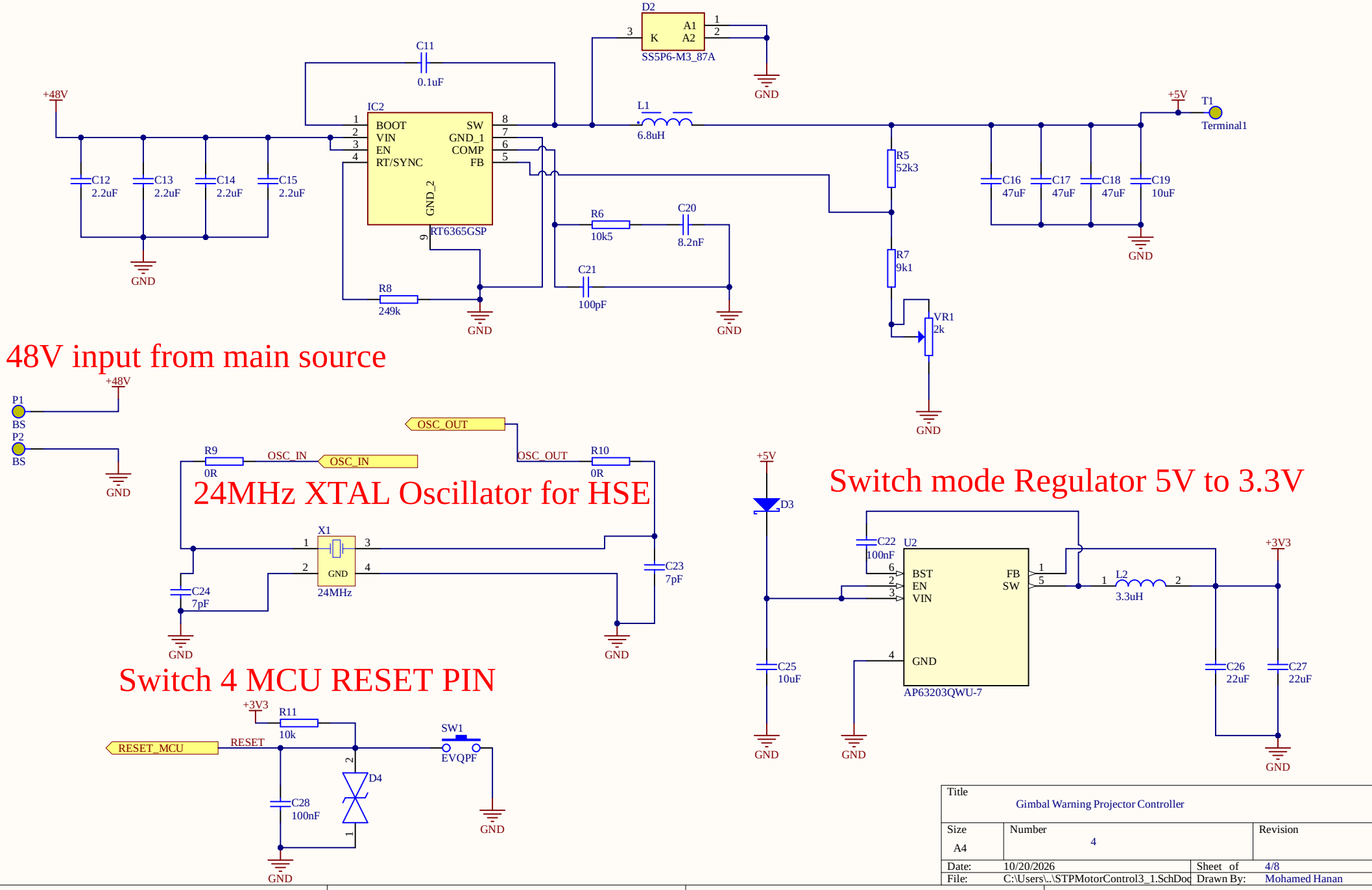


Title	Gimbal Warning Projector Controller	
Size: A4	Number: 2	Revision:
Date: 10/20/2026		Sheet of 2/8
File: C:\Users\..\STPMotorControl_1.SchDoc		Drawn By: Mohamed Hanan

CAN Transciever

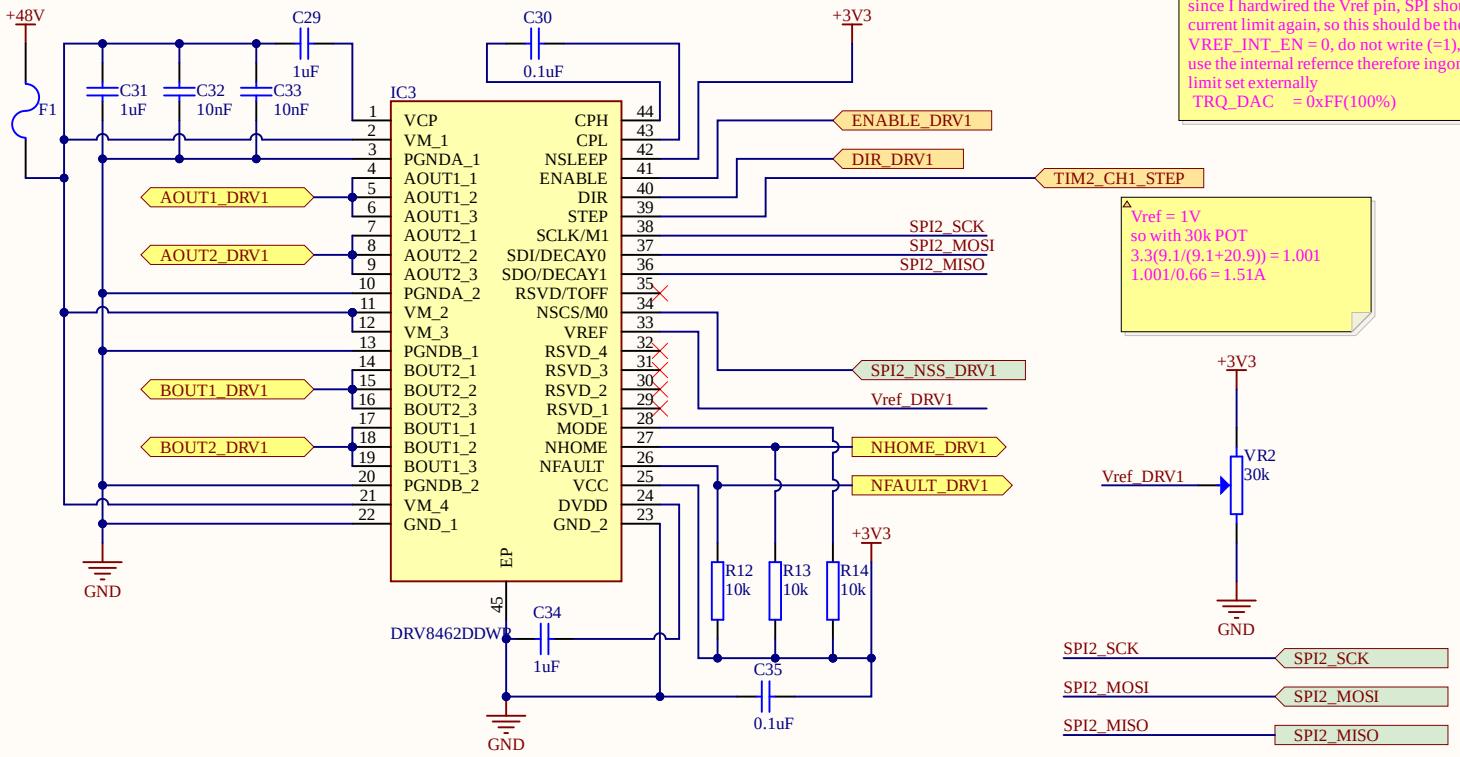


Title	Gimbal Warning Projector Controller		
Size: A4	Number: 3		Revision:
Date: 10/20/2026		Sheet of 3/8	
File: C:\Users\..\STPMotorControl2_1.SchDoc		Drawn By: Mohamed Hanan	



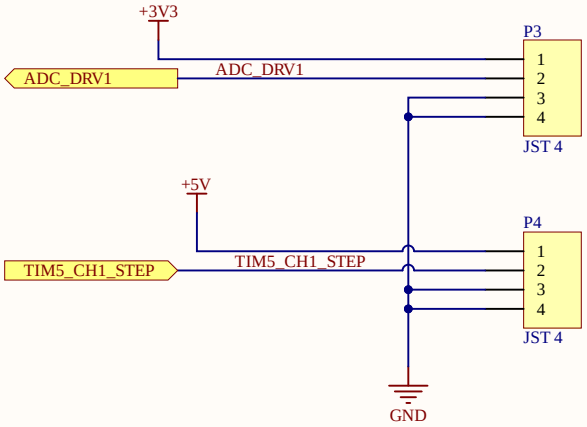
Title	Gimbal Warning Projector Controller	
Size: A4	Number: 4	Revision:
Date: 10/20/2026	Sheet of 4/8	
File: C:\Users\..\STPMotorControl3_1.SchDoc	Drawn By: Mohamed Hanan	

STEPPER DRIVER1



Title	Gimbal Warning Projector Controller	
Size: A4	Number: 5	Revision:
Date: 10/20/2026		Sheet of 5/8
File: C:\Users\..\STPMotorControl4_1.SchDoc		Drawn By: Mohamed Hanan

MAGNETIC ENCODER CONNECTOR

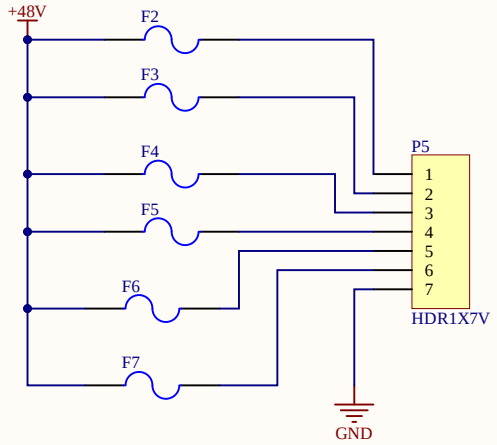


STP Port

25CH-B-04
female counter part
25CTB-R

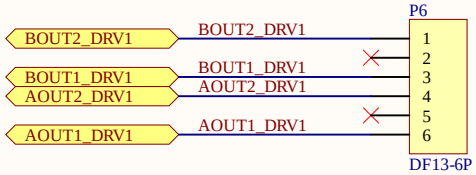
Servo Port

LED modules Connector

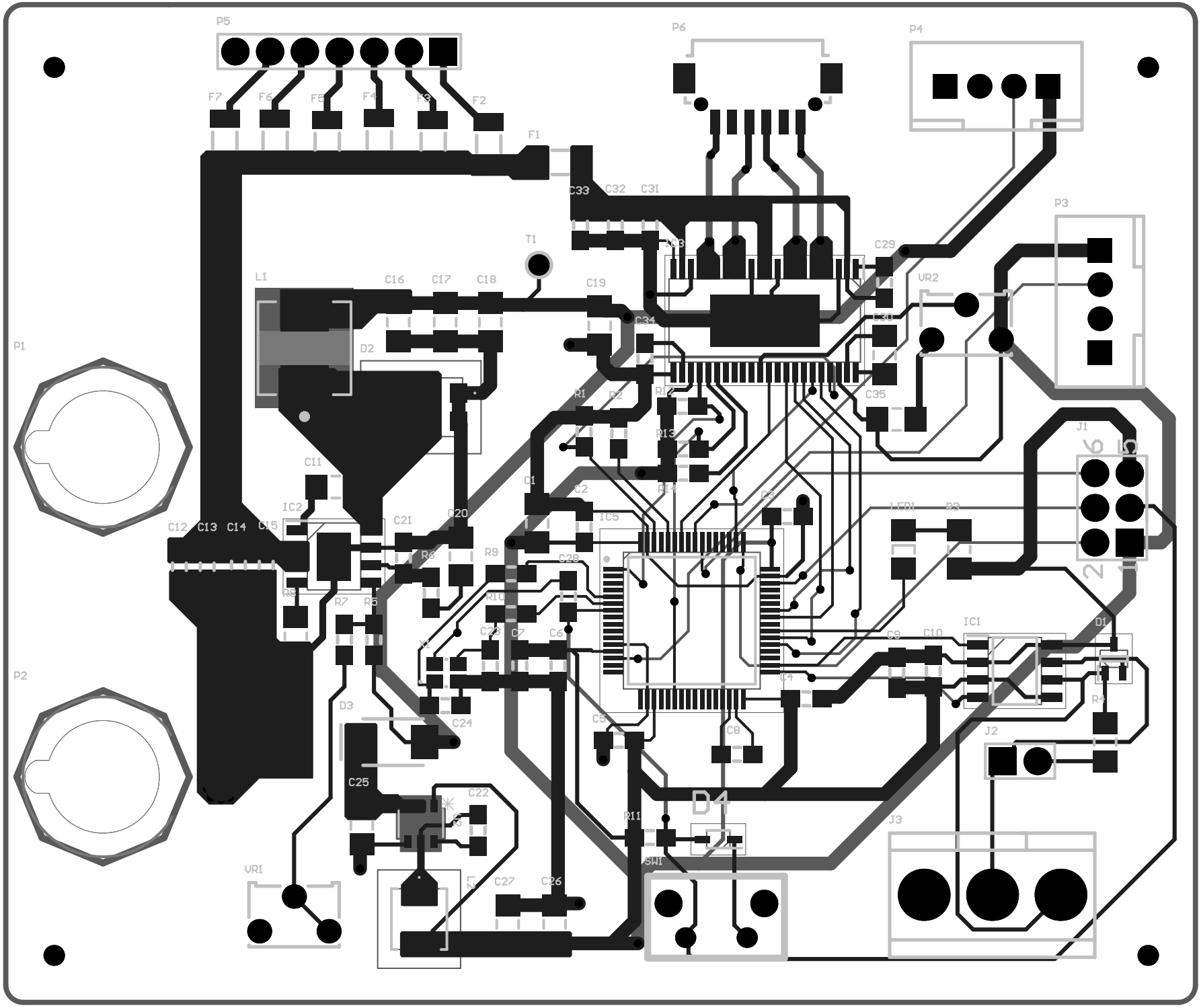


VHR-7N
female counter part
SVH-41T-P1.1

STEPPER MOTOR CONNECTOR



Title	Gimbal Warning Projector Controller		
Size: A4	Number: 8		Revision:
Date: 10/20/2026		Sheet of: 8/8	
File: C:\Users\..\STPMotorControl7_1.SchDoc		Drawn By: Mohamed Hanan	


P5
P6
P4
P3
P1
P2
J1
J2
J3
T1
L1
D2
D1
D3
D4
IC1
IC2
IC5
VR1
VR2
SW1
LED1
F1
F2
F3
F4
F5
F6
F7
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
C27
C28
C29
C30
C31
C32
C33
C34
C35
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
X1

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7uF	Non Polarized Capacitor, SMD, 1206	C1	CAP_1206	CAPN 1206	1
100nF	Non Polarized Capacitor, SMD, 0805	C2, C3, C4, C5, C7, C10, C22, C28	CAP_0805	CAPN 0805	8
1uF	Non Polarized Capacitor, SMD, 0805	C6, C9, C29, C31, C34	CAP_0805	CAPN 0805	5
2.2uF	Non Polarized Capacitor, SMD, 0805	C8, C12, C13, C14, C15	CAP_0805	CAPN 0805	5
0.1uF	Non Polarized Capacitor, SMD, 1206	C11, C30, C35	CAP_1206	CAPN 1206	3
47uF	Non Polarized Capacitor, SMD, 1206	C16, C17, C18	CAP_1206	CAPN 1206	3
10uF	Non Polarized Capacitor, SMD, 1206	C19, C25	CAP_1206	CAPN 1206	2
8.2nF	Non Polarized Capacitor, SMD, 1206	C20	CAP_1206	CAPN 1206	1
100pF	Non Polarized Capacitor, SMD, 0805	C21	CAP_0805	CAPN 0805	1
7pF	Non Polarized Capacitor, SMD, 0805	C23, C24	CAP_0805	CAPN 0805	2
22uF	Non Polarized Capacitor, SMD, 1206	C26, C27	CAP_1206	CAPN 1206	2
10nF	Non Polarized Capacitor, SMD, 0805	C32, C33	CAP_0805	CAPN 0805	2
ESDCAN03-2BWY	TVS Diode (Uni-directional)	D1	SOT65P210X110-3N	ESDCAN03-2BWY	1
SS5P6-M3_87A	Schottky Diode	D2	SS5P6M387A	SS5P6-M3_87A	1
PMEG3050BEP	Schottky REC, 30V, 5A, SMD	D3	SOD-128	PMEG3050BEP	1
TSD05CDYFR	TVS Diode (Bi-directional)	D4	SOD2613X100N	TSD05CDYFR	1
1206L035/16YR	Fuse, PTC Reset, 350mA Hold, 750mA Tripping, 16v	F1, F2, F3, F4, F5, F6, F7	MC36207	1206L035/16YR	7
TCAN3403DRQ1	Integrated Circuit	IC1	SOIC127P600X175-8N	TCAN3403DRQ1	1
RT6365GSP	Integrated Circuit	IC2	SOIC127P600X175-9N	RT6365GSP	1
DRV8462DDWR	Integrated Circuit	IC3	SOP64P810X120-45N	DRV8462DDWR	1
STM32C552RET6	Integrated Circuit	IC5	QFP50P1200X1200X160-64N	STM32C552RET6	1
HDR3X2	Header, 3-Pin, Dual row, 2.54mm Pitch, Straight	J1	HDR2X3_AUT	HDR3X2	1
HDR1X2V	Header, Plug, 2.54mm, 2 Pins, Straight	J2	HDR1X2_AUT	HDR1X2V	1
SCREWX3	Terminal Block, PCB, 3Way, Pitch 5mm	J3	SCREW3_AUT	SCREWX3	1
6.8uH	Power Inductor, 6.8Uh, Shielded, 6.6A Rohs Compliant: Yes | Wurth Elektronik 7443734948068	L1	WE-LHMI_7050	WE-LHMI_7050	1
3.3uH	Inductor	L2	SRP4020TA3R3M	SRP4020TA-3R3M	1
LED-RED	RED LED, Case 1206	LED1	LED1206R	SMD 1206 LED-RED	1
BS	Banana Socket, 4mm, Panel, Black, 10A	P1, P2	Banana_AUT	Banana Socket	2
JST4	JST XA Header, 4-Pin, Pitch 2.5mm	P3, P4	XA-4	JST XA Header 4	2
HDR1X7V	Header, Plug, 2.54mm, 7 Pins, Straight	P5	HDR1X7_AUT	HDR1X7V	1
DF13-6P	HIROSE(HRS) DF13-6P-1.25H(20) Wire-To-Board Connector, Right Angle, DF13 Series, Surface Mount, Header, 6 Contacts, 1.25 mm	P6	DF13-6	DF13-6P	1
5k1	Resistor, SMD, 0805	R1	RES_0805	Resistor 0805	1
10k	Resistor, SMD, 0805	R2, R11, R12, R13, R14	RES_0805	Resistor 0805	5
1k	Resistor, SMD, 1206	R3	RES_1206	Resistor 1206	1
120R	Resistor, SMD, 1206	R4	RES_1206	Resistor 1206	1
52k3	Resistor, SMD, 0805	R5	RES_0805	Resistor 0805	1
10k5	Resistor, SMD, 0805	R6	RES_0805	Resistor 0805	1
9k1	Resistor, SMD, 0805	R7	RES_0805	Resistor 0805	1
249k	Resistor, SMD, 1206	R8	RES_1206	Resistor 1206	1
0R	Resistor, SMD, 0805	R9, R10	RES_0805	Resistor 0805	2
EVQPF	PANASONIC EVQPF203M SWITCH, TACTILE, SPST-NO, 20mA, 15VDC	SW1	EVQPF203M	EVQPF	1
Terminal1	Terminal, Pad 70x70mil, Hole 32 mil	T1	PAD1_AUT	Terminal1	1
AP63203QWU-7	Description	U2	TSOT26_DIO	AP63203QWU-7	1
2k	3362S Series, 1/4 " Square Potentiometer, 2k, 10%, 0.5 Watts, 1Turn	VR1	3362S	3362S-1-202LF	1
30k	3362S Series, 1/4 " Square Potentiometer, 2k, 10%, 0.5 Watts, 1Turn	VR2	3362S	3362S-1-202LF	1
24MHz	Crystals, 24MHz	X1	CX2520SB	CX2520SB24000D0FLJZZ	1